Review summary: predicting fate and toxicity from structure and/or MS/MS spectra

Plata lab hackathon work on environmental fate and toxicity. Built 2026-11-01 (v8) from the same result files as the whitepaper; figure numbers refer to the whitepaper. This summary is generated from the paper's own numbers; it has not itself been independently audited, and the whitepaper marks section by section which results a second agent checked.

What we did

A swarm of AI agents built a predictor for 12 environmental fate endpoints and a set of toxicity tasks, and then audited it. The predictor accepts three kinds of input: a structure, a structure plus a measured MS/MS spectrum, or a spectrum alone. Each prediction comes with a conformal interval, a domain flag and the EPI Suite value for comparison. Accuracy is the RMSE as a percentage of a naive guess (always predicting the training mean; 100% = no skill). We report it first on leak-free test compounds, i.e. those with no training analogue at Tanimoto >= 0.40 on Morgan radius-2, 2048-bit fingerprints. That definition is fingerprint-dependent: 57.4% of leak-free test rows still share a Bemis-Murcko scaffold with a training compound of the same endpoint (`READER-1/split_leakage_audit.md [not redistributed; VM-internal]`), so the stricter novel-scaffold cut is reported in the whitepaper, Section 4.1. **The model that was frozen and that produces every number below is a structure + AIMNet2-physics model with no spectral features at all** (`W-MODEL/frozen_v5/columns.json [not redistributed; VM-internal]`: two blocks, 2,296 columns, zero spectral); the spectral results come from separate models and did not produce the headline or the EPI Suite comparison.

Bottom line

All of the following comes from one model and one file: the shipped structure+physics model, scored on identical leak-free rows against EPI Suite in `data/fate_vs_epi_suite.csv [shipped aggregate; working file not redistributed; VM-internal]`.

endpoint	n leak-free	ours, % of naive (leak-free)	EPI Suite, same rows	EPI - ours, pp (observed) [95% CI]	label noise floor	interval coverage, leak-free (nominal)
vapor_pressure	302	45%	93%	+47.6 [+11.4, +80.5]	27%	0.90 (0.90)
water_solubility	403	50%	87%	+37.0 [+23.6, +49.7]	30%	0.91 (0.90)
logkow	451	51%	93%	+41.3 [+0.3, +92.6]	37%	0.87 (0.90)
koc	27	52%	65%	+12.9 [+0.1, +25.6]	-	0.85 (0.90)
henry	121	52%	77%	+24.5 [-3.3, +53.3]	28%	0.96 (0.90)
melting_point	820	53%	84%	+31.2 [+25.7, +37.2]	22%	0.90 (0.90)
boiling_point	367	53%	78%	+25.5 [+8.1, +43.1]	28%	0.93 (0.90)
koh_gas	30	58%	27%	-31.0 [-51.0, -15.8]	-	0.93 (0.90)
pka_basic	54	74%	no EPI module	-	-	0.89 (0.90)
bcf	52	88%	99%	+11.1 [-10.2, +44.5]	45%	0.92 (0.90)
pka_acidic	38	97%	no EPI module	-	-	0.95 (0.90)
logkoa	3	n<20	-	-	-	n<20 (0.90)

Skill (95% CI upper bound below 100% of naive, `data/headline_leakfree.csv`, variant SP) on leak-free compounds: boiling_point, henry, koc, koh_gas, logkow, melting_point, pka_basic, vapor_pressure, water_solubility. No skill: pka_acidic. BCF is 88% of naive with a 95% CI of [68, 104], which crosses 100, so skill is not established for BCF on leak-free compounds (W-MODEL peer check). Too few leak-free compounds to score: logkoa.

Spectrum only (Fig. 3)

From one HR ESI+ spectrum and no structure (W-SPECTRA's single-spectrum model, which is not the shipped fate model), leak-free, as % of naive with 95% CI: water_solubility 74.7 [67.0, 82.4], vapor_pressure 70.3 [56.4, 87.0], logkow 78.9 [70.1, 87.6], melting_point 86.1 [79.3, 93.1]; skill is not established for boiling_point (82.6 [67.7, 106.3]) or henry (92.9 [77.8, 103.5]), and the pKa endpoints, Koc and BCF have fewer than 20 leak-free ESI+ spectra and are not scored leak-free. We quote

per-endpoint values rather than a pass/fail list: melting point clears the 85%-of-naive bar on W-MODEL's pooled spectra-only ablation arm and misses it on the single-spectrum model, by 1.1 points over on the single-spectrum arm and 4.0 points under on the pooled arm, on row sets that differ (n=198 vs n=218); on the 198 identical rows the two arms differ by 1.4 points of naive with a 95% cluster-bootstrap CI of [-3.26, +6.32] that includes zero, so it is shown and not claimed (`W-SPECTRA/spectrum_only_final_table.md [not redistributed; VM-internal]`). [DISPUTED, deliberately left open: the spectrum-only skill count is 3 or 4 depending on which model is quoted - W-SPECTRA's single-spectrum model puts melting point at 86.1% of naive leak-free (HR ESI+), over the 85% bar, and W-MODEL's pooled spectra-only ablation arm puts it at 81.0%, under it (`W-SPECTRA/spectrum_only_final_table.md [not redistributed; VM-internal]`). Melting point is shown, not promoted.] These intervals under-cover (empirical 0.81-0.87 against a nominal 0.90). Fragments beat the precursor mass for henry, koc, logkow, melting_point, pka_acidic, vapor_pressure, water_solubility; a known structure is better for henry, logkow, melting_point, vapor_pressure, water_solubility. The rebuilt spectral representation did not beat the Run-2 one (better for none, worse for none): a clean negative result for that hypothesis.

Physics and spectra as add-ons (Fig. 2)

SP: leak-free gain with CI excluding zero on boiling_point, melting_point, pka_acidic, pka_basic. No feature block flattens the rise in error with unfamiliarity. We report this as a negative result.

Intervals (Fig. 4)

Nominal 90%; empirical coverage 0.81-1.00 across endpoint x familiarity cells (full test, n>=20). Three cells fall below nominal on a Wilson interval, one of the three (boiling_point >=0.7) on the Clopper-Pearson interval the result file reports, and leak-free logKow at nn_sim 0.3-0.5 is a fourth (0.863, n=336): boiling_point 0.5-0.7 0.87 (n=540); boiling_point >=0.7 0.83 (n=93); koc 0.3-0.5 0.81 (n=53). Coverage holds per familiarity bin but not per chemical class (ionizable classes under-cover).

Hazard endpoints, on measured labels

Three hazard tasks were rebuilt on measured labels (ChEMBL dose-response for receptor potency, CompTox for rat oral LD50); the Run-2 receptor numbers, which scored against another model's consensus calls, are withdrawn. As % of the naive training-mean guess on leak-free compounds: rat oral LD50 87% (n=1012), oestrogen-receptor potency 52% (n=255), androgen-receptor potency 83% (n=149). For the rare-positive activity calls we report PR-AUC against each familiarity bin's own base rate, never accuracy: in the least familiar bin the oestrogen-receptor model reaches 0.329 against a base rate of 0.047 (precision 0.24, recall 0.44). A control that knows only the similarity bin already reaches PR-AUC ar_active 0.606, er_active 0.515, so much of the pooled ranking skill is prevalence drift rather than chemistry.

What the tool faces outside the test split

We scored the two product questions over the EPA inventory rather than the test split: a 40,000-structure random sample for definedness and familiarity, and 635,818 inventory structures through the shipped model (of 649,665 with physics descriptors at the time the prediction pass was cut; the relaxation campaign itself reached 782,199 structures). 10% of the predicted inventory is below Tanimoto 0.3 to anything we trained on and 7% is above 0.7 - the reverse of a random test split. Up to 4.0% of the inventory is physically undefined for the gas-phase endpoints, about four times its share of the labelled data, and a further 7-16% is measurable but not equivalent to our labels; neither is a data gap. Re-weighting the hazard models by the inventory shows the test split was flattering androgen-receptor potency, and that the oestrogen-receptor intervals under-cover for the typical inventory chemical. One open problem, stated plainly: at equal Tanimoto similarity the inventory gets far wider intervals than the test set, so a similarity bin does not fully describe how far the inventory has moved from our training data.

What failed, measured cleanly

- **Physics is a small, unstable effect.** The set of endpoints where AIMNet2 descriptors beat structure alone with a CI excluding zero changed with every defensible scoring choice (v4: henry, vapour pressure; v5 pre-de-duplication: pka_acidic, pka_basic; v5 de-duplicated: boiling_point, melting_point, pka_acidic, pka_basic), and no feature block flattens the rise of error with unfamiliarity.

- **Rebuilding the spectral representation did not rescue spectra.** Under the same model and the same compounds the rebuilt representation is no better than the Run-2 one; the old representation was not the problem. Spectra also add nothing measurable on top of a known structure.
- **The spectral domain flag does not predict error.** Median absolute rank correlation between error and the best-matching training spectrum's cosine similarity is 0.063 across 10 endpoints.
- **The model does not know transformation chemistry.** It predicts parent-to-product changes no better than changes between arbitrary similar compounds matched on size and similarity.
- **Spectrum-only fails for acute lethality, and spectra add nothing on top of a structure** (and the frozen model that produced every fate number contains no spectral features at all). The full spectrum reaches 95.8% of naive against 81.6% from the structure on the same 345 compounds - a wide interval around the training mean - and structure + spectrum minus structure spans zero. This is a well-powered negative result on the central question of the work, and it is in the main text of the paper, not a footnote.
- **Our label-fidelity bet (P-004) is falsified for the androgen receptor and not established for ER-alpha.** Models trained on measured receptor potency transferred worse to an external literature benchmark than models trained on another model's consensus labels - but the published pair compared a continuous-potency arm against a binary-label arm on a binary benchmark, which handicaps the measured arm. On the matched arm (measured ChEMBL *call*) the differences are ER-alpha [-0.117, +0.009], spanning zero, and AR [-0.193, -0.084], not spanning zero. Consensus labels are also smoother than pooled measurements, so the winning arm may have the more *learnable* labels rather than the more *correct* ones; we cannot separate those tonight.
- **Our own definedness bookkeeping was wrong for one compound class, and an expert cross-check caught it.** A standardisation step neutralised sulfonate and sulfate salts before the rule that decides whether a property exists was applied, so 6 gas-phase and neutral-species cells covering 678 skeletons were called defined when they are physically undefined; 251 labelled rows are affected. The table was rebuilt, the shipped predictor now refuses those cells, and a related SMARTS defect that gave sodium dodecyl sulfate a basic pKa of 6.6 was found and fixed in the dashboard (`CRITIC/undefined_vs_gap_cells_q003.csv`, `dashboard/tool/wtool_postproc.py`).
- **Acidic pKa has no skill on novel chemistry** and EPI Suite's AOPWIN beats us on the gas-phase OH rate constant (n=30 leak-free rows, all of them PhysProp-sourced), and we do not know why. Both are reported rather than hidden.
- **Tooling that failed:** a vendor AIMNet2 wrapper silently dropped the Coulomb term without Ewald (found and documented by OPS-2); the mechanism-matched open-shell jobs could not run because of a GPU driver/warp incompatibility; an enviPath-based transformation arm could not be built without network access; and logKoa has three leak-free test compounds, too few to score at all.

Data priorities

These come from a leave-class-out experiment with a size-matched control, not from counting empty cells: a cell can be empty because the property is physically undefined for that class, and those cells are excluded. The largest measured novelty penalties, in points of naive error, are sulfonic/sulfate acid or salt for water_solubility (+95 pp, CI [31.0, 144.8], n=30); aromatic hydrocarbon for koc (+44 pp, CI [32.6, 57.3], n=80); sulfonic/sulfate acid or salt for logkow (+43 pp, CI [33.9, 51.6], n=25); nitroaromatic/nitro for henry (+34 pp, CI [4.2, 94.5], n=19); aromatic hydrocarbon for bcf (+28 pp, CI [13.6, 44.0], n=138); halogenated aliphatic for bcf (+28 pp, CI [5.8, 55.8], n=44). A learning-curve estimate on the same data shows that extra random measurements help familiar chemistry most, so untargeted data collection will not close these gaps; measurements inside the named classes will.

Decisions for the team

1. **Ship the fate endpoints, gate the hazard ones.** Four fate endpoints - melting point (n=820), water solubility (403), boiling point (367) and vapour pressure (302) - beat EPI Suite on compounds with no training analogue, and they are the four that survive a Bonferroni correction over the nine leak-free comparisons that were made (`data/fate_vs_epi_suite.csv [shipped aggregate; working file not redistributed; VM-internal]`). logKow and Koc are better unadjusted but not after correction, and we do not claim them; Henry's law and BCF are not distinguishable; EPI Suite is decisively better on the gas-phase OH rate constant (27% vs our 58% of naive, n=30) and we do not explain that away. [DISPUTED, deliberately left open: how far the EPI 'home-turf' caveat extends. On PhysProp rows intersected with leak-free, melting point and water solubility keep a margin over EPI Suite and boiling point, vapour pressure, logKow, Henry and BCF are not resolved (`data/label_source_attack.csv`);

on ALL PhysProp test rows vapour pressure also keeps one (`data/dg2_fairness_dedup.csv`). The two cells disagree; a human should pick. See whitepaper Section 4.1.] Receptor activity should be presented as a ranked screen against a stated base rate, never as a call. 2. **Keep the undefined/missing distinction in the product.** Refusing to predict a gas-phase property for a permanently charged species is a feature; counting those cells as data gaps would misdirect measurement money. 3. **Treat spectrum-only as a per-endpoint capability.** It has skill for logKow, water solubility and vapour pressure; melting point is shown, not claimed [DISPUTED, deliberately left open: the spectrum-only skill count is 3 or 4 depending on which model is quoted - W-SPECTRA's single-spectrum model puts melting point at 86.1% of naive leak-free (HR ESI+), over the 85% bar, and W-MODEL's pooled spectra-only ablation arm puts it at 81.0%, under it (`W-SPECTRA/spectrum_only_final_table.md [not redistributed; VM-internal]`). Melting point is shown, not promoted.]; it does not work for acute lethality, and the intervals under-cover on novel chemistry and need recalibration before release. 4. **Do not invest further in single-conformer physics descriptors** without the conformer-ensemble version; the seed conformer is close to a random draw for flexible molecules. 5. **Fund measurements by class, not by count** - the classes listed above, and a replicate study for BCF, whose label noise floor is the highest we measured.